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Earphone

Abstract

An earphone is provided, and includes a housing, a loudspeaker, and a noise cancellation microphone, the housing comprises an internal space, and a sound pickup channel. The noise cancellation microphone is connected to the communication surface by the sound pickup channel. The housing comprises an outer housing and a frame, the housing is provided with the internal space and a communication hole, and the frame is provided with a communication surface facing the communication hole. The communication surface comprises a sound-emitting hole and a sound pickup hole, the sound-emitting hole and the sound pickup hole are isolated from each other, and the sound-emitting hole is connected to the front cavity. The sound pickup channel and the front cavity are isolated from each other through the frame.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national stage of International Application No. PCT/CN2021/139404, filed on Dec. 18, 2021, which claims priority to Chinese Patent Application No. 202011612556.8, filed on Dec. 30, 2020. Both of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(2) This application relates to the earphone field, and in particular, to an earphone.

BACKGROUND

(3) Currently, earphones in the market usually adopt an active noise cancellation mode to cancel noise heard by human ears, to optimize sound effect and sound quality of the earphones. Currently, there are mainly three active noise cancellation modes: feedforward active noise cancellation, feedback active noise cancellation, and hybrid active noise cancellation. The feedforward active noise cancellation means that noise of an outside world is picked up and the noise in the environment is cancelled. In this case, noise cancellation effect for wind noise is poor, and even the wind noise is amplified. The feedback active noise cancellation means that noise in an ear canal is picked up, and for a headphone or an in-ear earphone, passive noise cancellation effect of an ear tip and an ear cup is good, and impact of wind noise is small. Therefore, a noise cancellation microphone disposed in the ear tip and the ear cup has good noise cancellation effect. The hybrid active noise cancellation is implemented in combination with the foregoing two noise cancellation modes. For a semi-in-ear earphone, because there is no passive noise cancellation structure like an ear tip or an ear cup, a feedback noise cancellation microphone is interfered by an outside world, resulting in poor noise cancellation effect.

SUMMARY

(4) In view of this, an earphone needs to be provided to improve noise cancellation effect.

(5) According to a first aspect of embodiments of this application, an earphone is provided. The earphone includes a housing, a speaker, a noise cancellation microphone, a dust filter, and a cosmetic mesh. The housing is provided with inner space, a communication surface, and a sound pickup channel. The communication surface is configured to communicate the inner space with an outside world, a sound-emitting hole is provided on the communication surface, and the sound-emitting hole communicates with the inner space. The speaker is disposed in the inner space, the sound pickup channel is formed with a sound pickup opening on the communication surface, and the sound pickup channel is further formed with a sound pickup hole at another location of the housing. The noise cancellation microphone is disposed on the housing and covers the sound pickup hole, the dust filter is disposed on the communication surface, and the cosmetic mesh is disposed on the housing and covers the communication surface.

(6) In the earphone, the sound-emitting hole is designed on the communication surface, the sound pickup opening is designed on the communication surface through the sound pickup channel, and the dust filter and the cosmetic mesh are added. In this way, the noise cancellation microphone receives noise toward an ear canal through the sound pickup opening, so that the received noise can be as similar as possible to noise in the ear canal. This optimizes a condition for noise cancellation debugging, and achieves better noise cancellation effect.

(7) In a possible design of the first aspect, the dust filter covers the sound-emitting hole and the sound pickup opening. The sound-emitting hole and the sound pickup opening are covered, so that better dustproof effect can be achieved.

(8) In a possible design of the first aspect, a sound resistance of an area of the dust filter that covers the sound pickup opening is lower than a sound resistance of an area of the dust filter that covers the sound-emitting hole. A lower sound resistance of the sound pickup opening helps the noise cancellation microphone receive noise through the sound pickup opening, so that the received noise can be as similar as possible to the noise in the ear canal.

(9) In a possible design of the first aspect, a sound resistance of the dust filter at the sound pickup opening is less than or equal to 10 MKS Rayls.

(10) In a possible design of the first aspect, the dust filter covers the sound-emitting hole, but does not cover the sound pickup opening. The sound pickup opening is not covered, so that the noise cancellation microphone can receive the noise in the ear canal at maximum.

(11) In a possible design of the first aspect, the housing includes an outer housing and a frame. The outer housing is provided with a communication hole and the inner space, and the communication hole communicates with the outside world. The frame is disposed in the inner space and is disposed close to the communication hole, the frame is provided with the communication surface facing the

communication hole, and the sound-emitting hole communicates the inner space with the communication hole. The sound pickup channel is disposed on the frame, the noise cancellation microphone is disposed on the frame, and the sound pickup opening communicates the sound pickup channel with the communication hole. The cosmetic mesh is located between the outer housing and the frame and covers the communication hole. The frame is designed to facilitate reasonable designs of locations of the sound pickup channel and the sound-emitting hole.

(12) In a possible design of the first aspect, the outer housing includes a front housing and a rear housing. The front housing is disposed on the rear housing, the front housing and the rear housing jointly enclose the inner space. The frame and the speaker are disposed on the front housing, the speaker and the frame divide the inner space into a front cavity and a rear cavity, the front cavity is located between the speaker and the frame, and the rear cavity is located between the speaker and the rear housing. The noise cancellation microphone is located in the front cavity, the communication hole is disposed on the front housing, and the communication hole communicates the front cavity with the outside world. The dust filter and the cosmetic mesh are disposed between the front housing and the frame. The front and rear housings are designed to facilitate mounting of the frame and the speaker, and facilitate production and processing.

(13) In a possible design of the first aspect, the frame is provided with a mounting surface close to the sound-emitting hole. The sound pickup hole is formed on the mounting surface and is close to the communication surface, the noise cancellation microphone is disposed on the mounting surface through a connecting part, and the sound pickup hole communicates with a sound pickup end of the noise cancellation microphone. A design of the mounting surface close to the sound-emitting hole is conducive to achieving a maximum approach of the noise cancellation microphone to the communication surface, and is more conducive to receiving the noise in the ear canal.

(14) In a possible design of the first aspect, the earphone further includes a sealing part, and the sealing part is configured to seal the outer housing and the frame at the communication hole and the communication surface. The sealing part is designed to protect against dust and prevent a sound wave from spreading through a gap.

(15) In a possible design of the first aspect, the sealing part includes a first sealing part, a second sealing part, and a third sealing part. The first sealing part is disposed between the dust filter and the communication surface, to seal the dust filter on the communication surface. The second sealing part is disposed between the outer housing and the cosmetic mesh, so that the cosmetic mesh is disposed on the outer housing, and the communication hole is sealed. The third sealing part is disposed between the dust filter and the cosmetic mesh, to implement sealing between the dust filter and the cosmetic mesh.

(16) In a possible design of the first aspect, each of the first sealing part and the third sealing part is provided with openings at respective locations relative to the sound-emitting hole and the sound pickup opening, so that the sound-emitting hole and the sound pickup opening are separated at the communication hole between the outer housing and the frame. In this way, a sound emitted by the speaker from the sound-emitting hole has minimal impact on the noise cancellation microphone in the sound pickup opening, so that the noise cancellation microphone can receive the noise in the ear canal as much as possible.

(17) In a possible design of the first aspect, the first sealing part, the second sealing part, and the third sealing part are double-sided adhesives or glue. This saves costs and improves mounting efficiency.

(18) In a possible design of the first aspect, the sound pickup opening is isolated from the sound-emitting hole, and the sound pickup channel is isolated from the front cavity through the frame. This minimizes impact of the speaker on sound pickup of the noise cancellation microphone.

(19) In a possible design of the first aspect, the rear housing includes an earphone front end and an earphone rear end. The earphone front end is used for mounting the speaker and the front housing, and the earphone rear end is used for hand holding. The earphone rear end is convenient for a user

to hold, so that the earphone is worn on an ear of the user.

(20) In a possible design of the first aspect, the rear cavity is provided with a circuit board, a battery module, a power input unit, a communication unit, or a call microphone.

(21) In a possible design of the first aspect, the cosmetic mesh is a metal mesh. This improves style and mechanical reliability of the earphone, and prevents the inner dust filter, speaker, and noise cancellation microphone from being damaged by an external force.

(22) In a possible design of the first aspect, a sound resistance of the cosmetic mesh is less than or equal to 10 Rayls MKS.

(23) In a possible design of the first aspect, the sound pickup opening is located on a side that is of the communication surface and that is away from the speaker. This helps reduce impact of the speaker on sound pickup of the noise cancellation microphone.

(24) In a possible design of the first aspect, the earphone is further provided with an active noise cancellation microphone. This can further improve sound quality.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of a structure of an earphone according to a first embodiment of this application;

(2) FIG. 2 is a partial exploded three-dimensional schematic diagram of the earphone shown in FIG. 1;

(3) FIG. 3 is a three-dimensional schematic diagram of a frame of the earphone shown in FIG. 2;

(4) FIG. 4 is a sectional view of the earphone shown in FIG. 1;

(5) FIG. 5 is a sectional view of an earphone according to a second embodiment of this application;

(6) FIG. 6 is a sectional view of an earphone according to a third embodiment of this application; and

(7) FIG. 7 is a sectional view of an earphone according to a fourth embodiment of this application.

DESCRIPTION OF REFERENCE NUMERALS FOR MAIN COMPONENTS

(8) Earphone **100, 200, 300** Housing **10** Outer housing **11** Front housing **111** Rear housing **112** Earphone front end **1121** Earphone rear end **1122** Frame **12** Communication surface **121** Sound-emitting hole **122** Sound pickup channel **123** Sound pickup opening **124** Sound pickup hole **125** Mounting surface **126** Inner space **13** Front cavity **131** Rear cavity **132** Communication hole **14** Speaker **20** Noise cancellation microphone **30** Connecting part **31** Dust filter **40** Cosmetic mesh **50** Sealing part **60** First sealing part **61** Second sealing part **62** Third sealing part **63**

(9) In the following specific embodiments, the present invention is further described with reference to the accompanying drawings.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(10) The following describes the technical solutions in embodiments of this application with reference to the accompanying drawings in embodiments of this application.

(11) The following terms “first”, “second”, and the like are merely used for description, and shall not be understood as an indication or implication of relative importance or implicit indication of a quantity of indicated technical features. Therefore, a feature limited by “first” or “second” may explicitly or implicitly include one or more features. In the descriptions of this application, unless otherwise stated, “a plurality of” means two or more than two. Orientation terms such as “up”, “down”, “left”, and “right” are defined relative to an orientation of schematic placement of components in the accompanying drawings. It should be understood that these directional terms are relative concepts and are used for relative description and clarification. These directional terms may vary accordingly depending on an orientation in which the components are placed in the accompanying drawings.

(12) In this application, unless otherwise explicitly specified and limited, a term “connection” should be understood in a broad sense. For example, the “connection” may be a fastened connection, a detachable connection, an integrated connection, may be a direct connection, or an indirect connection through an intermediate medium. The term “and/or” used in this specification includes any and all combinations of one or more related listed items.

(13) When the following embodiments are described in detail with reference to schematic diagrams, for ease of description, a diagram indicating a partial structure of a component is partially enlarged not based on a general scale. In addition, the schematic diagrams are merely examples, and should not limit the protection scope of this application herein.

Embodiment 1

(14) FIG. 1 and FIG. 2 show an earphone **100** according to a first embodiment of this application. The earphone **100** can perform active noise cancellation to cancel noise in an ear canal, to improve sound effect and sound quality of the earphone **100**. The earphone **100** may be an in-ear earphone, a semi-in-ear earphone, a headphone, or the like, and in particular, relates to a semi-in-ear earphone.

(15) The earphone **100** includes a housing **10**, a speaker **20**, a noise cancellation microphone **30**, a dust filter **40**, and a cosmetic mesh **50**. The housing **10** is provided with inner space **13** that communicates with the outside. The speaker **20** and the noise cancellation microphone **30** are disposed in the inner space **13** (refer to FIG. 4) of the housing **10**. The dust filter **40** and the cosmetic mesh **50** are disposed on the housing **10** and are located at a location at which the inner space **13** communicates with the outside, so that the inner space **13** communicates with the outside through the dust filter **40** and the cosmetic mesh **50**.

(16) The housing **10** includes an outer housing **11** and a frame **12**. The outer housing **11** is provided with the inner space **13** and a communication hole **14**. The communication hole **14** is configured to communicate the inner space **13** with an outside world. In addition, when the earphone **100** is worn on a human ear, the communication hole **14** points to an ear canal of a human body. The frame **12** is disposed in the inner space **13** of the outer housing **11** and is disposed close to the communication hole **14**. The noise cancellation microphone **30** is disposed on the frame **12**. The dust filter **40** and the cosmetic mesh **50** are located between the outer housing **11** and the frame **12** and cover the communication hole **14**.

(17) Specifically, the outer housing **11** includes a front housing **111** and a rear housing **112**. The front housing **111** is disposed on the rear housing **112**, and the front housing **111** and the rear housing **112** jointly enclose the inner space **13**. The frame **12** is disposed on the front housing **111**. The speaker **20** is disposed on the front housing **111**. The speaker **20** and the frame **12** divide the inner space **13** into a front cavity **131** and a rear cavity **132**. The front cavity **131** is located between the speaker **20** and the frame **12**. The rear cavity **132** is located between the speaker **20** and the rear housing **112**. The noise cancellation microphone **30** is located in the front cavity **131**. The communication hole **14** is disposed on the front housing **111**. The communication hole **14** communicates the front cavity **131** with the outside world. The dust filter **40** and the cosmetic mesh **50** are disposed between the front housing **111** and the frame **12** and cover the communication hole **14**.

(18) As shown in FIG. 3 and FIG. 4, an inner wall of the frame **12** roughly matches an inner wall of the front housing **111**. The frame **12** is provided with a communication surface **121** facing the communication hole **14**. A sound-emitting hole **122** is provided on the communication surface **121**. The sound-emitting hole **122** communicates the front cavity **131** with the communication hole **14**. The frame **12** is further provided with a sound pickup channel **123**. The sound pickup channel **123** is formed with a sound pickup opening **124** on the communication surface **121**, and the sound pickup channel **123** is further formed with a sound pickup hole **125** at another location of the frame **12**. The noise cancellation microphone **30** is disposed on the frame **12** and covers the sound pickup hole **125**. On the communication surface **121**, the sound pickup opening **124** is isolated from the sound-emitting hole **122**, and the sound pickup channel **123** is isolated from (not communicates

with) the front cavity **131** through the frame **12**, to minimize impact of the speaker **20** on sound pickup of the noise cancellation microphone **30**.

(19) In this embodiment, the sound pickup opening **124** is located on a side that is of the communication surface **121** and that is away from the speaker **20**. There are two sound-emitting holes **122**, and the sound-emitting holes **122** are symmetrically disposed on the communication surface **121**. The sound pickup opening **124** is located between the two sound-emitting holes **122**. However, this is not limited. In another embodiment, there may alternatively be one or more sound-emitting holes **122**.

(20) The speaker **20** is configured to emit a sound to the front cavity **131**, to transmit the sound to the ear canal through the sound-emitting hole **122** and the communication hole **14**.

(21) The noise cancellation microphone **30** is a feedback microphone (Feedback Mic, FB Mic, or Error Mic), and is configured to receive, pick up, and detect residual noise in an ear canal between the earphone and a human ear after the earphone is worn, and enable, by using an algorithm (a built-in unit of the noise cancellation microphone **30**, another unit in the earphone **100**, or an external processing unit), the speaker **20** of the earphone to release a reverse sound wave to cancel the noise.

(22) An inner wall of the frame **12** is provided with a mounting surface **126** close to the sound-emitting hole **122**. The sound pickup hole **125** is formed on the mounting surface **126** and is close to the communication surface **121**. The noise cancellation microphone **30** is disposed on the mounting surface **126** through a connecting part **31**, and the sound pickup hole **125** communicates with a sound pickup end of the noise cancellation microphone **30**. In this way, noise in the ear canal is transmitted to the noise cancellation microphone **30** through the communication hole **14** and the sound pickup channel **123**.

(23) The connecting part **31** may be an adhesive like a double-sided adhesive or glue.

(24) As shown in FIG. 4, the dust filter **40** is disposed on the communication surface **121** and covers the sound-emitting hole **122** and the sound pickup opening **124**. The cosmetic mesh **50** is disposed on the front housing **11** and completely covers the communication hole **14**.

(25) The earphone **100** further includes a sealing part **60**. The sealing part **60** is configured to seal the outer housing **11** and the frame **12** at the communication hole **14** and the communication surface **121**.

(26) The sealing part **60** includes a first sealing part **61**, a second sealing part **62**, and a third sealing part **63**. Both the dust filter **40** and the cosmetic mesh **50** are planar meshes whose shapes are similar to a shape of the communication surface **121**. The first sealing part **61** is disposed between the dust filter **40** and the communication surface **121**, to seal the dust filter **40** on the communication surface **121**. The second sealing part **62** is disposed between the front housing **11** and the cosmetic mesh **50**, so that the cosmetic mesh **50** is disposed on the front housing **111**, and the communication hole **14** is sealed. The third sealing part **63** is disposed between the dust filter **40** and the cosmetic mesh **50**, to implement sealing between the dust filter **40** and the cosmetic mesh **50**.

(27) Each of the first sealing part **61** and the third sealing part **63** is provided with openings at respective locations relative to the sound-emitting hole **122** and the sound pickup opening **124**, so that the sound-emitting hole **122** and the sound pickup opening **124** are separated at the communication hole **14** between the outer housing **11** and the frame **12**. In this way, the sound emitted by the speaker **20** from the sound-emitting hole **122** has minimal impact on the noise cancellation microphone **30** in the sound pickup opening **124**, so that the noise cancellation microphone **30** can receive the noise in the ear canal as much as possible. The second sealing part **62** is substantially ring-shaped, so that the cosmetic mesh **50** is disposed on a wall surface around the communication hole **14** of the front housing **111**.

(28) Adhesives such as a double-sided adhesive or glue may be selected for the first sealing part **61**, the second sealing part **62**, and the third sealing part **63**.

(29) The dust filter **40** is configured to prevent external dust from entering the earphone and affecting sound effect and sound quality of the earphone.

(30) The cosmetic mesh **50** is preferably a metal mesh, to improve style and mechanical reliability of the earphone, and prevent the inner dust filter **40**, speaker **20**, and noise cancellation microphone **30** from being damaged by an external force, for example, prevent an external sharp object from piercing.

(31) A damping of the cosmetic mesh **50** needs to be as small as possible. Preferably, a sound resistance of the cosmetic mesh **50** is less than 10 Rayls MKS (Rayleigh, where in an MKS system, 1 Rayleigh is 1 Pascal-second per meter, and 1 Rayls=1 Pa×s/m), to ensure that the cosmetic mesh **50** has small impact on sound pickup of the noise cancellation microphone **30**.

(32) The rear housing **112** includes an earphone front end **1121** and an earphone rear end **1122**. The earphone front end **1121** is used for mounting the speaker **20** and the front housing **11**. The earphone rear end **1122** is convenient for a user to hold, so that the earphone **100** is worn on an ear of the user.

(33) During use, the user holds the earphone rear end **1122** and wears, on the ear, the earphone front end **1121** provided with the front housing **11**, so that a side provided with the communication hole **14** faces the ear canal. The noise cancellation microphone **30** receives noise in an ear canal, and the speaker **20** emits a sound with a reverse sound wave, so that the noise in the ear canal is eliminated from a sound heard by the human ear, thereby improving sound effect and sound quality of the earphone.

(34) The rear cavity **132** is configured to dispose another component of the earphone, for example, a circuit board configured to control the earphone, a battery module configured to store electric energy, a power input unit configured to communicate with an external power supply unit, a communication unit configured to perform signal transmission with the outside, or a call microphone configured to receive a sound made by the user.

(35) Further, the earphone **100** may be provided with a component and design that improve sound quality and wearing comfort in a common earphone, like an active noise cancellation microphone (Feedforward Mic or Ref Mic) or a pressure relief hole.

(36) According to the earphone **100** provided in this application, the communication surface **121** faces the ear canal, the independent sound pickup channel **123** is disposed on the frame **12** to enable the sound pickup opening **124** to be as close as possible to the ear canal, and the sound pickup opening **124** of the sound pickup channel **123** and the sound-emitting hole **122** are separated on the communication surface **121**. In addition, the dust filter **40** is added, and the sound pickup opening **124** and the sound-emitting hole **122** between the communication hole **14** and the communication surface **121** are separated by the sealing part **60**. In this way, the noise cancellation microphone **30** receives noise toward the ear canal through the sound pickup opening **124**, so that the received noise can be as similar as possible to the noise in the ear canal. This optimizes a condition for noise cancellation debugging, achieves better noise cancellation effect, and achieves better dustproof effect.

Embodiment 2

(37) FIG. 5 shows an earphone **200** according to a second embodiment of this application.

(38) The earphone **200** has a same structure as the earphone **100**, and includes the housing **10**, the speaker **20**, the noise cancellation microphone **30**, the dust filter **40**, and the cosmetic mesh **50**. A main difference from the earphone **100** in the first embodiment shown in FIG. 1 lies in that structures of the dust filter **40**, the cosmetic mesh **50**, and the sealing part **60** are different.

(39) Specifically, the dust filter **40** covers only the sound-emitting hole **122**, but does not cover the sound pickup opening **124**. The cosmetic mesh **50** is in an outwardly protruding shape. The third sealing part **63** in the sealing part **60** is substantially ring-shaped, to connect an outer periphery of the cosmetic mesh **50** to an outer periphery of the dust filter **40** in a sealed manner.

(40) According to the earphone **200** provided in this application, the communication surface **121**

faces the ear canal, the independent sound pickup channel **123** is disposed on the frame **12**, and the sound pickup opening **124** of the sound pickup channel **123** and the sound-emitting hole **122** are separated on the communication surface **121**. In this way, the noise cancellation microphone **30** receives noise toward the ear canal through the sound pickup opening **124**, so that the received noise can be as similar as possible to the noise in the ear canal. This optimizes a condition for noise cancellation debugging, and achieves better noise cancellation effect.

Embodiment 3

(41) FIG. **6** shows an earphone **300** according to a third embodiment of this application.

(42) The earphone **300** has a same structure as the earphone **200**, and includes the housing **10**, the speaker **20**, the noise cancellation microphone **30**, the dust filter **40**, and the cosmetic mesh **50**. A main difference from the earphone **200** in the second embodiment shown in FIG. **5** lies in that a structure of the dust filter **40** is different.

(43) Specifically, aperture density of an area of the dust filter **40** that covers the sound-emitting hole **122** is different from aperture density of an area of the dust filter **40** that covers the sound pickup opening **124**. Therefore, an aperture of the dust filter **40** at the sound pickup opening **124** is larger, and in this case, a sound resistance of the dust filter **40** at the sound pickup opening **124** is lower (less than or equal to 10 MKS Rayls).

(44) According to the earphone **300** provided in this application, the communication surface **121** faces the ear canal, the independent sound pickup channel **123** is disposed on the frame **12**, and the sound pickup opening **124** of the sound pickup channel **123** and the sound-emitting hole **122** are separated on the communication surface **121**. In addition, the dust filter **40** is added, and a damping of the dust filter **40** at the sound pickup opening **124** is reduced through adjustment. In this way, the noise cancellation microphone **30** receives noise toward the ear canal through the sound pickup opening **124**, so that the received noise can be as similar as possible to the noise in the ear canal. This optimizes a condition for noise cancellation debugging, and achieves better noise cancellation effect.

Embodiment 4

(45) FIG. **7** shows an earphone **400** according to a fourth embodiment of this application.

(46) The earphone **400** has a same structure as the earphone **100**, and includes the housing **10**, the speaker **20**, the noise cancellation microphone **30**, and the dust filter **40**. A main difference from the earphone **100** in the second embodiment shown in FIG. **1** lies in that the cosmetic mesh **50** is not provided.

(47) Specifically, the dust filter **40** may be made of a material with high mechanical strength, to simultaneously implement functions of the dust filter **40** and the cosmetic mesh **50** in the earphone **100**. The sealing part **60** includes only the first sealing part **61** and the second sealing part **62**. The first sealing part **61** is disposed between the dust filter **40** and the communication surface **121**, to seal the dust filter **40** on the communication surface **121**. The second sealing part **62** is disposed between the front housing **111** and the dust filter **40**, so that the dust filter **40** is disposed on the front housing **111**, and the communication hole **14** is sealed.

(48) Similar to that in the earphone **300** in Embodiment 3, aperture density on the dust filter **40** in the earphone **400** may further be designed as that an aperture for covering the sound-emitting hole **122** is greater than an aperture for covering the sound pickup opening **124**. In this case, a sound resistance of the dust filter **40** at the sound pickup opening **124** is lower.

(49) Certainly, in another embodiment, there may be a plurality of dust filters **40**, and the dust filters **40** separately cover the sound-emitting hole **122** and the sound pickup opening **124** (may be disposed on a hole wall or on a surface of the communication surface **121**). The sealing part **60** includes only the second sealing part **62**, and the second sealing part **62** is disposed between the front housing **11** and the communication surface **121**, to seal the communication hole **14** on the front housing **11** through the communication surface **121**.

(50) According to the earphone **400** provided in this application, the communication surface **121**

faces the ear canal, the independent sound pickup channel **123** is disposed on the frame **12**, and the sound pickup opening **124** of the sound pickup channel **123** and the sound-emitting hole **122** are separated on the communication surface **121**. In addition, the dust filter **40** is added, and a damping of the dust filter **40** at the sound pickup opening **124** is reduced through adjustment. In this way, the noise cancellation microphone **30** receives noise toward the ear canal through the sound pickup opening **124**, so that the received noise can be as similar as possible to the noise in the ear canal. This optimizes a condition for noise cancellation debugging, and achieves better noise cancellation effect.

(51) This application further provides a terminal apparatus, where the terminal apparatus has the earphone **100**, the earphone **200**, or the earphone **300**.

(52) The terminal apparatus may be glasses, an AR apparatus, a VR apparatus, or another product with audio effect.

(53) The foregoing description is merely specific implementations of this application, but is not intended to limit the protection scope of this application. Any variation or replacement within the technical scope disclosed in this application shall fall within the protection scope of this application.

Claims

1. An earphone, comprising a housing, a loudspeaker, and a noise cancellation microphone, wherein: the housing comprises an internal space, a communication surface for connecting the internal space and an outside world, and a sound pickup channel; the noise cancellation microphone is connected to the communication surface by the sound pickup channel; the loudspeaker is disposed in the internal space, the internal space comprises a front cavity, and the loudspeaker is connected to the communication surface by the front cavity; the housing comprises an outer housing and a frame, the housing is provided with the internal space and a communication hole, and the frame is provided with the communication surface facing the communication hole; the communication surface comprises a sound-emitting hole and a sound pickup hole, the sound-emitting hole and the sound pickup hole are isolated from each other, and the sound-emitting hole is connected to the front cavity; the sound pickup channel is disposed on the frame, the sound pickup channel forms a radio reception port on the communication surface and forms a sound reception hole at a position of the frame, and the sound reception hole is connected to a sound reception end of the noise cancellation microphone; and, the sound pickup channel and the front cavity are isolated from each other through the frame.

2. The earphone according to claim 1, wherein the noise cancellation microphone is disposed on the frame.

3. The earphone according to claim 1, wherein the housing comprises a front housing and a rear housing, and the front housing and the rear housing jointly surround to form an inner space, and wherein at least one of: the frame is disposed on the front housing; or the loudspeaker is disposed on the front housing.

4. The earphone according to claim 3, wherein the rear housing comprises a front end of the earphone and a rear end of the earphone, the front end of the earphone is configured to install the loudspeaker and set the front housing, and the rear end of the earphone is configured to be handheld.

5. The earphone according to claim 1, wherein the internal space comprises the front cavity and a rear cavity; wherein the front cavity is located between the loudspeaker and the frame; and wherein the rear cavity is located between the loudspeaker and a rear housing.

6. The earphone of claim 5, wherein the noise cancellation microphone is located in the front cavity.

7. The earphone according to claim 5, wherein the rear cavity is provided with a circuit board, a

battery module, a power input unit, a communications unit, or a call microphone.

8. The earphone according to claim 1, wherein the frame is disposed in the internal space of the housing, and wherein the frame is disposed close to the communication surface.

9. The earphone according to claim 1, wherein a sound reception port is located on a connection surface, and the sound reception port is located on a side of a sound output hole away from the loudspeaker.

10. The earphone according to claim 9, wherein the earphone further comprises a dust filter, and a dustproof net is disposed on the communication surface.

11. The earphone according to claim 10, wherein the dust filter covers the sound-emitting hole and the sound pickup hole.

12. The earphone according to claim 10, wherein the earphone further comprises an appearance net, and the appearance net covers the connection surface and the dust filter.

13. The earphone according to claim 12, wherein the appearance net is a metal mesh.

14. The earphone according to claim 12, wherein a shape of the appearance net is an outwardly protruding shape.

15. The earphone according to claim 12, wherein the earphone further comprises a seal, and the seal is configured to seal the housing and the frame at the communication hole and the communication surface.

16. The earphone according to claim 15, wherein the seal comprises a first seal, a second seal, and a third seal, and the first seal is disposed between the dust filter and the communication surface, so as to seal the dustproof net on the communication surface; wherein the second seal is disposed between the housing and the appearance net, so that the appearance net is disposed on the housing and the communication hole is sealed; and wherein the third seal is disposed between the dustproof net and the appearance net, so as to seal the dustproof net and the appearance net.

17. The earphone according to claim 16, wherein an opening is disposed on both the first seal and the third seal relative to the sound-emitting hole and the sound pickup hole; and wherein the sound-emitting hole at the communication hole and the sound pickup hole are separated between the housing and the frame.

18. The earphone according to claim 17, wherein the first seal, the second seal, and the third seal are double-sided adhesive or glue.

19. The earphone according to claim 1, wherein a mounting surface is disposed on the frame close to a sound output hole, the sound receiving hole is formed on the mounting surface and close to the communication surface, and the noise cancellation microphone is disposed on the mounting surface by a connector.
